

ML-Driven Fraud Detection System

Dataset Critical Analysis & Preparation Report

Executive Summary

Dataset Size	11,400 transactions × 26 features
Fraud Cases	997 frauds (8.75% rate)
Legitimate Cases	10,403 (9125.44%)
Missing Values	Up to 305 per column (~2.7%)
Duplicate Rows	200 exact duplicates
Date Range	0000-00-00 to 2025/13/40

Critical Data Quality Issues

The dataset contains multiple data quality issues that must be addressed before model training:

Missing Values: Up to 305 missing values across various features (KYC tier, fees, device trust scores, IP data)

Duplicates: 200 exact duplicate rows require deduplication

Type Mismatches: amount_src stored as string instead of numeric

Text Inconsistencies: Whitespace, case variations, and typos in categorical columns (channel, kyc_tier, home_country)

Outliers: Extreme values in fees (\$9,999.99), amounts (\$12,498.57), and negative fees (-\$1.00)

Class Imbalance: Only 8.75% fraud rate requiring specialized handling

Data Cleaning & Preparation Steps

Phase 1: Data Loading & Initial Assessment

1. Load dataset with UTF-8 encoding
2. Document data dictionary and column descriptions
3. Identify and correct data type misclassifications

Phase 2: Handle Missing Values

1. Drop rows with missing timestamp (29 rows) - critical for temporal analysis
2. Impute KYC tier (300 missing) using mode or 'missing' category
3. Impute IP-related fields (305 missing) as 'unknown' or forward-fill
4. Impute fee (295 missing) using median or 0

5. Impute amount_usd (305 missing) using exchange rate calculation or median

Phase 3: Remove Duplicates

1. Identify and remove 200 exact duplicate rows
2. Validate that duplicates are not legitimate multi-leg transactions

Phase 4: Fix Data Type Issues

1. Convert amount_src from string to float - critical for calculations
2. Parse timestamp to datetime format with timezone handling
3. Ensure categorical columns have proper object dtype

Phase 5: Standardize Text & Fix Inconsistencies

1. Standardize home_country: strip whitespace, uppercase, consolidate: { ' UK ', ' US ', ' CA ' } → {UK, US, CA}
2. Standardize channel: lowercase, fix typos (mobile→mobile, weeb→web, ATm→ATM)
3. Standardize kyc_tier: lowercase, fix typos (standrd→standard, enhancd→enhanced), consolidate nulls

Phase 6: Fix Data Inconsistencies

1. Validate exchange rates are positive and within reasonable bounds
2. Fix negative fees - set to 0 or investigate refunds
3. Validate all currency codes against ISO 4217 standards
4. Reconcile source vs destination currencies - flag self-transfers

Phase 7: Handle Outliers

1. Cap amount_usd outliers at 99th percentile or log-transform
2. Fix extreme fee values (\$9,999.99) - likely data entry errors
3. Analyze account_age_days outliers - very new accounts show higher fraud
4. Validate transaction velocity outlier distributions

Phase 8: Feature Engineering & Validation

1. Create source-destination currency pair feature (27 unique pairs)
2. Validate derived risk scores are within appropriate bounds
3. Verify boolean columns contain only True/False
4. Validate chargeback_history_count is non-negative

Phase 9: Address Class Imbalance

1. Document fraud distribution: 8.75% fraud rate (997 vs 10,403)
2. Plan for SMOTE, class weights, or stratified sampling
3. Use stratified train-test-validation splits

Phase 10: Final Quality Checks

1. Validate temporal sequence of transactions
2. Check for data leakage (risk_score_internal calculation)
3. Detect multicollinearity among features
4. Create comprehensive data quality report

Data Quality Issues Priority Matrix

Issue	Impact	Priority
Missing values (~2.7%)	Data loss, potential bias	HIGH
200 duplicate rows	Biased statistics, data leakage	HIGH
Text inconsistencies	Models treat identical values differently	HIGH
String amount fields	Feature engineering impossible	HIGH
Extreme outliers	Skew model training	MEDIUM
Negative fees	Illogical values, likely errors	MEDIUM
Class imbalance (8.75%)	Model bias toward legitimate	MEDIUM
Timestamp inconsistencies	Velocity features unreliable	MEDIUM

Feature Overview

Feature	Type	Missing	Unique Values / Range
transaction_id	String	0	11,400
customer_id	String	0	Various
timestamp	DateTime	29	2022-10-03 to 2022-10-31
home_country	Categorical	0	7 (US, CA, UK, unknown, +duplicates)
amount_usd	Numeric	305	\$7.23 - \$12,498.57
amount_src	String→Numeric	0	9,856 unique values
fee	Numeric	295	-\$1.00 - \$9,999.99
channel	Categorical	0	12 (web, mobile, ATM, +duplicates)
kyc_tier	Categorical	300	14 unique including nulls/typos
account_age_days	Numeric	0	1 - 1,095 days
new_device	Boolean	0	True/False
ip_risk_score	Numeric	0	0 - 1
is_fraud (Target)	Binary	0	0 (10,403) / 1 (997)

Recommendations

Immediate Actions:

1. Execute all Phase 1-5 steps sequentially to ensure data integrity
2. Create before/after validation reports to track data quality improvements
3. Document all transformation decisions for reproducibility

Model Training Preparation:

4. Implement stratified train-test-validation splits (70-15-15) after deduplication
5. Apply SMOTE or class weights to address 8.75% fraud rate imbalance
6. Reserve 10% of fraud cases for final validation to prevent data leakage

Feature Engineering:

7. Create interaction features: (amount_usd × ip_risk_score), (new_device × location_mismatch)
8. Derive temporal features from timestamp: hour, day_of_week, day_of_month patterns
9. Encode categorical variables using target encoding for fraud correlation

Validation & Monitoring:

10. Implement robust error handling to flag future data quality issues
11. Create data quality dashboards to monitor drift in production
12. Establish data quality SLAs before model deployment